

 Do number, aur un ka farq

Atomic number (Z) = nucleus mein **protons** ki tadaad.

Ye har element ke liye **mukhtalif** hai, aur yehi element ki **shanakht** aur periodic table mein us ki **jagah** tay karta hai.

Mass number / Nucleon number (A) = **protons + neutrons**.

 Neutrons nikalne ka formula

Neutrons = A – Z

Misal: sodium ka A = 23 aur Z = 11. To neutrons = 23 – 11 = **12**.

Aur neutral atom mein electrons = protons = Z = **11**.

 Kaunsa number element badalta hai

Agar **protons** badal jayein to **element hi badal jata hai**. Agar sirf **neutrons** badlein to element wohi rehta hai, bas us ka **isotope** ban jata hai. Yehi agle safhay ka moazoo hai.

 Radioactivity

Jo isotopes **ghair mustahkam** hon wo **radioactive decay** se guzarte hain, yani toot kar doosra element ban jate hain.

Kitab ki do misalein: **C-14 → N-14** aur **U-238 → Pb-206**.

 Quiz

1. Define **atomic number** and **mass number**. An atom has A = 35 and Z = 17: how many protons, neutrons and electrons does it have?
2. What is **radioactive decay**? Give the two examples your book provides.